

Explicit Decoupled Memory for Sequential Agents

Evidence consolidation v0

This artifact consolidates the controlled evidence currently reproducible in the Embryo repository for explicit, decoupled memory in sequential agents.

Sized Claim

An explicit decoupled memory store can be written, learned, retrieved, and injected into a policy decision to improve controlled-task recall accuracy and retrieval efficiency on tasks where correct stored content is required.

- explicit memory recovers information beyond a measured recurrent baseline falloff;
- a learned associative reader recovers the hand-coded memory ceiling;
- recalled content can drive an action-level decision;
- recalled location content can drive a goal-conditioned navigator, making retrieval cheaper as the number of candidate locations grows.

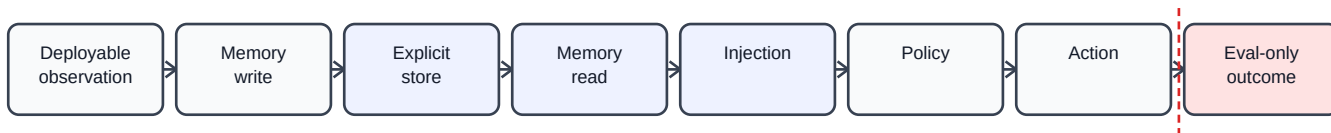
Non-Claims

- no learned write selection or full episodic memory management;
- no portability to external frozen backbones;
- no cross-episode memory;
- no world-like perception or high-dimensional visual grounding;
- no reinforcement-learning reward improvement;
- no Crafter competence or general embodied-agent competence.

Methodological Spine

- Fair baselines: task-trained recurrence or systematic search, not blind floors.
- Corruption controls: shuffled, wrong-binding, stale, content-corrupt, or order-corrupt arms.
- Crossover or scaling: effect appears where the baseline fails or grows with the controlled variable.
- Evaluability gates: contamination checks and reachability before effect interpretation.
- Anti-selection: tuning and final reporting stay separated.

Mechanism Overview



Actor/query inputs stop before eval-only outcome fields.

Eval-only fields score outcomes; they are not actor, query, or router inputs.

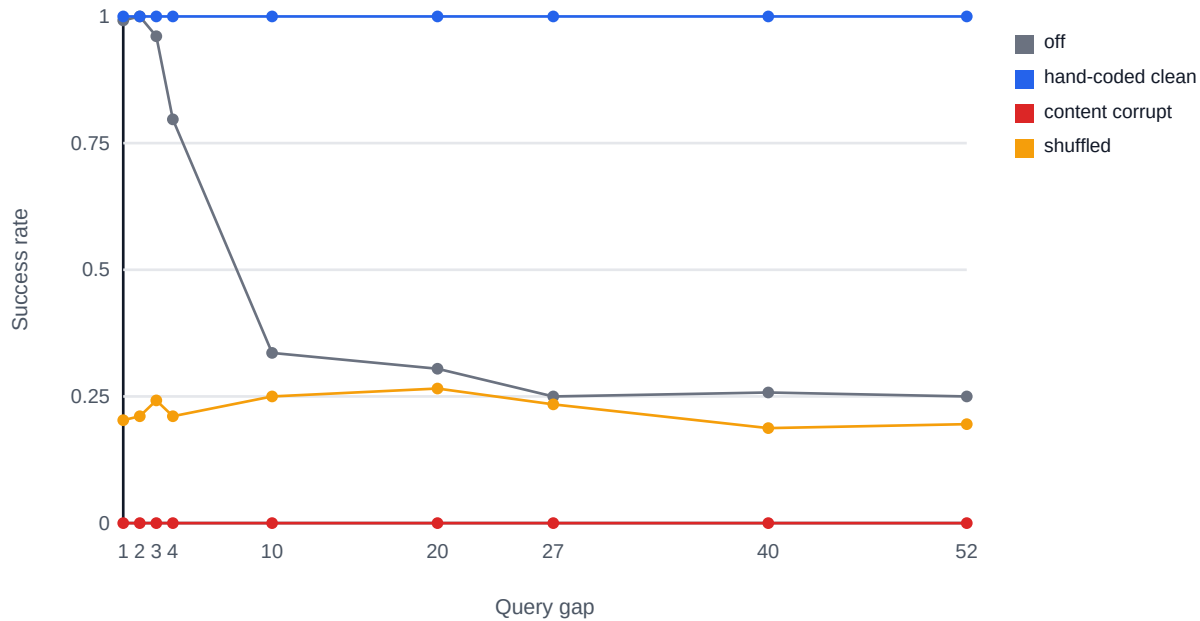
Evidence Rung 1: Recall Past Recurrent Saturation

Question: Can explicit memory recall information after a fair recurrent baseline falls off?

Task: POPGym Autoencode Easy.

Source artifacts: runs/popgym_autoencode_recurrent/popgym_autoencode_summary.json and runs/popgym_autoencode_recurrent/memory_curve.json.

Autoencode recall success by gap



Gap	No memory	Hand-coded clean	Content corrupt	Shuffled
1	0.992188	1	0	0.203125
4	0.796875	1	0	0.210938
10	0.335938	1	0	0.25
27	0.25	1	0	0.234375
52	0.25	1	0	0.195312

Sample count: 128 evaluated sequences per arm and gap.

Supports: explicit memory fills a measured recurrence gap. Does not support learned write selection or task performance beyond recall itself.

Evidence Rung 2: Learned Associative Read

Question: Can a learned memory reader recover the hand-coded ceiling?

Source artifacts: runs/popgym_autoencode_recurrent/learned_memory_curve.json and runs/popgym_autoencode_recurrent/learned_memory_checkpoint/manifest.json.

Autoencode learned-read success by gap



Arm	Success
Hand-coded clean	1
Learned clean	1
No memory	0.572049
Content corrupt	0
Order corrupt	0.248264
Shuffled	0.235243

Sample count: 1152 evaluated rows per aggregate arm.

Supports: the read can be learned and preserves the corruption signature. Does not support learned memory management; the write rule still stores the needed sequence.

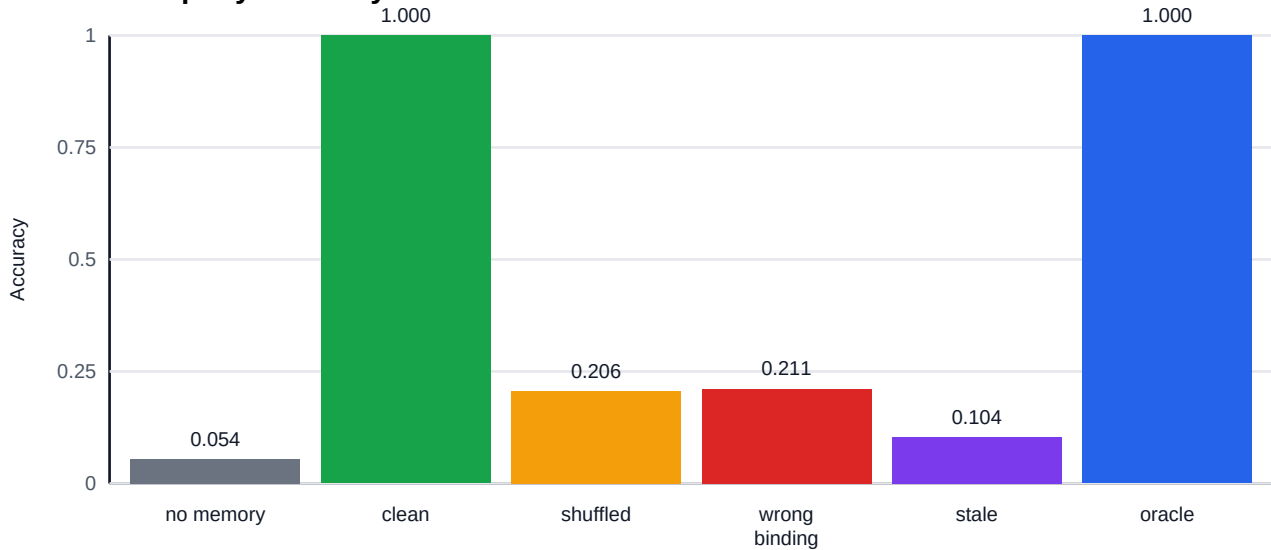
Evidence Rung 3: Action-Level Injection

Question: Can recalled content be injected into a decision when the correct action is the recalled value?

Task: POPGym CountRecall Hard.

Source artifacts: runs/popgym_count_recall_injection/popgym_count_recall_summary.json and runs/popgym_count_recall_injection/metrics_by_arm.json.

CountRecall query accuracy



Arm	Query accuracy (95% CI)	Mean reward	Invalid action rate
No memory	0.053744 +/- 0.003914	n/a	n/a
Clean memory	1 +/- 0	n/a	n/a
Shuffled	0.206295 +/- 0.010051	n/a	n/a
Wrong binding	0.211277 +/- 0.010061	n/a	n/a
Stale	0.103865 +/- 0.006417	n/a	n/a
Eval-only oracle	1 +/- 0	n/a	n/a

Named deviation: CountRecall has an integer-division/action-space off-by-one. The installed source sets `Discrete(max_card_count)`, but the rewarded count can equal `max_card_count`. Source receipt: `CountRecallHard Discrete(16)`, seed 12000, tick 191, target 16, `action_space.contains(target)=False`, `reward_for_target=0.00483092`. The probe corrects the local declared action space with `count_recall_inclusive_max_count_action_space_v0`; all arms use the same corrected space.

Supports: recalled content can cross from memory into a policy decision. Does not support goal-conditioned injection; here the action is the recalled count.

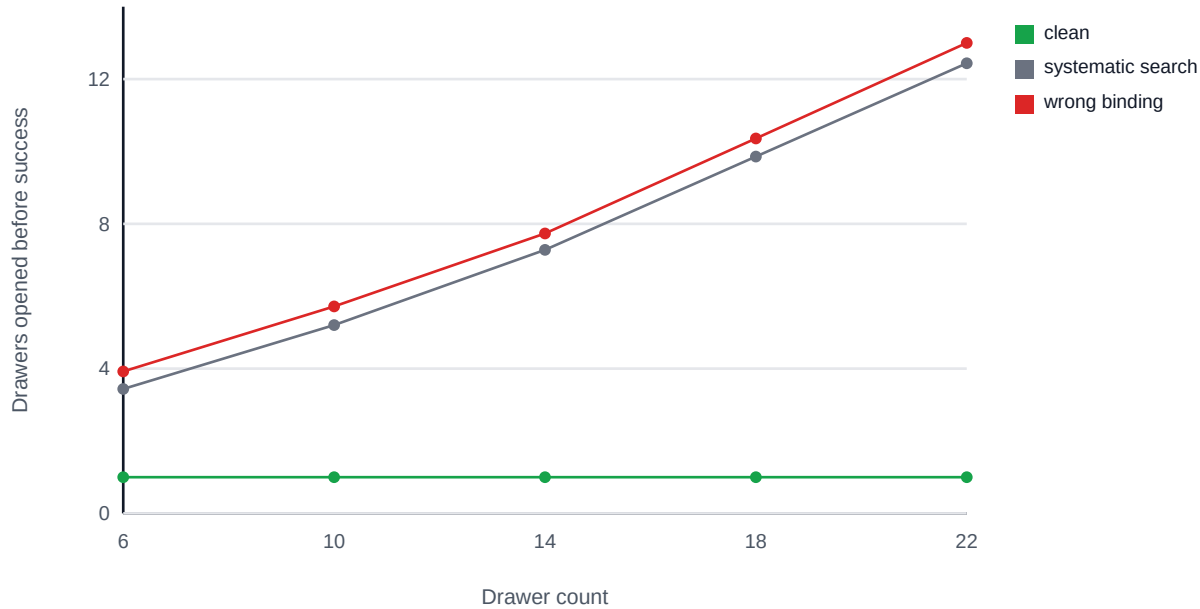
Evidence Rung 4: Goal-Channel Injection

Question: Can recalled content become a navigation goal for a policy whose actions are computed over multiple steps?

Task: controlled gridworld kitchen abstraction.

Protocol: reveal object location, write it to agent-owned memory, occlude, query the needed object, inject the recalled drawer as a goal into a frozen goal-conditioned navigator. If a memory-directed drawer is wrong, the agent falls back to systematic sweep.

Gridworld retrieval cost by drawer count



Arm	Success	Drawers opened	Steps	Wrong drawer
Systematic search	1	7.64375	39.0562	0.909375
Clean memory	1	1	5.54375	0
Wrong binding	1	8.14687	42.2875	1
Eval-only oracle	1	1	5.54375	0

Arm	Drawers opened (95% CI)	Episodes
Systematic search	7.64375 +/- 0.590433	320
Clean memory	1 +/- 0	320
Wrong binding	8.14687 +/- 0.567644	320

Clean and oracle drawer counts have structural zero variance because correct goal recall opens exactly one drawer.

Drawer count	Clean drawers	Search drawers	Gap
6	1	3.4375	2.4375
10	1	5.20312	4.20312
14	1	7.28125	6.28125
18	1	9.85938	8.85938
22	1	12.4375	11.4375

Supports: goal-channel injection works when recall and action are separated by the policy's path computation. The memory advantage grows with search-space size. Aggregate wrong-binding step cost over systematic search is 3.23125 steps because the navigator commits to the wrong recalled goal before fallback.

Does not support learned write selection, cross-episode memory, real perception, or external-backbone portability.

Negative and Scoped-Out Diagnostics

- Crafter long-run work supports memory-evaluability and mechanism diagnostics, not Crafter competence.
- Closed-loop reward probes showed that behavior-cloned recurrent baselines can have high per-step teacher agreement while failing task completion.
- Runnable maze-like tasks are not automatically memory-necessary tasks.
- A tuned scalar bias can create apparent survival gains on a small evaluation set; those numbers are not memory evidence without held-out selection discipline.

Current Boundary

The consolidated evidence is strongest for controlled within-episode associative recall and injection efficiency.

It is not evidence yet for external-backbone portability, learned write selection, cross-episode memory, reinforcement-learning reward improvement, or world-like embodied competence.

Next Keystone Choice

Desired claim	Required next step
Injection efficiency	Review this controlled result.
External-backbone portability	Scout and reproduce a competent external goal-conditioned backbone, then measure adapter cost and shuffle-gated me
Full episodic memory	Build a distractor-heavy task where storing everything fails and a learned write-selection rule is required.
Reward-performance improvement	Build or borrow a fair RL baseline first.

Reproduction Commands

```
python3 scripts/build_evidence_consolidation_figures.py
python3 scripts/build_evidence_consolidation_pdf.py
python3 -m compileall -q embryo scripts tests examples
python3 -m unittest discover -s tests
python3 scripts/check_release.py
```